Viraj Induruwa

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PROFILE SUMMARY

Backend Software Engineer specializing in Python, with hands-on experience building and deploying production-grade APIs, business logic engines, and financial calculation systems for enterprise clients. Currently developing and maintaining a **Sales Incentive Automation System** at Sri Lanka Telecom, including purpose-built Python console applications for each incentive module, automating complex financial calculations that previously required days of manual processing. Strong foundation in FastAPI, Django, PostgreSQL, Oracle Database, and Linux server deployment, with practical experience integrating secure authentication systems (JWT, OAuth, Azure AD) and cloud infrastructure (AWS, Azure). Proven track record of automating complex business workflows and delivering measurable impact in live production environments.

WORK EXPERIENCE

Freelance Software Engineer Oct 2025 - Present
Sri Lanka Telecom (SLT) Colombo, Sri Lanka

- Continued development and enhancement of the **Sales Incentive Automation System** on SLT's internal infrastructure, deployed on **Linux servers** with **Azure AD** authentication and **Oracle DB** integration
- Led the development of a new **Manager Incentive Module**, automating managerial-level incentive calculations with advanced business logic
- Currently developing two additional automated calculation services to extend the incentive automation platform's coverage across further business workflows
- Transitioned business workflows from manual Excel and Tableau-based processing to a centralized automated system, reducing reporting turnaround time from **7 days to under 3 minutes**
- Enhanced operational efficiency by **90%** while significantly reducing human calculation errors

Intern Software Engineer Apr 2025 - Oct 2025
Sri Lanka Telecom (SLT) Colombo, Sri Lanka

- Contributed to the development of the **Sales Incentive Automation System**, replacing manual Excel-based workflows used by sales and dealer management teams
- Designed and implemented a **Python-based console application** to automate complex sales incentive and dealer commission calculations
- Developed backend APIs using **Django and FastAPI**, integrating with **PostgreSQL and Oracle Database**
- Implemented secure authentication and role-based access control using **Azure Active Directory (Azure AD), Google OAuth, and JWT**
- Built **Python-based data processing pipelines** for validation, transformation, and automation of business data used in incentive calculations
- Improved calculation accuracy and reduced manual processing time by approximately **90%**

SKILLS

- **Programming Languages:** Python, JavaScript, SQL
- **Backend & API Development:** FastAPI, Flask, Django, RESTful API Design, JWT, OAuth, Microservices, System Design, Authentication & Authorization
- **Databases & Data Engineering:** PostgreSQL, Oracle Database, MySQL, SQLite, Database Schema Design, Pandas, NumPy, ETL Concepts
- **Frontend:** React, Tailwind CSS

- **Cloud, DevOps & Tools:** AWS (EC2, Lambda, S3, RDS), familiar with Azure Active Directory and Google Cloud Platform (GCP), Docker, Linux, Git, GitHub, Bitbucket, Postman, Playwright
- **Virtualization:** Oracle VirtualBox, familiar with VMware vSphere and Hyper-V concepts
- **AI Engineering & Automation Tools:** Claude Code, OpenAI Codex, GitHub Copilot, VS Code with Copilot, ChatGPT, Microsoft Copilot; applied AI tools to accelerate scripting, debugging, documentation, and automation workflows
- **AI / Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-Learn, Hugging Face Transformers, OpenCV, YOLO, Tesseract OCR, RAG Systems, LLM Integration
- **Professional Skills:** Analytical Thinking, Problem Solving, Attention to Detail, Team Collaboration, Communication, Leadership, Adaptability, Project Management, Time Management

PROJECTS

LoopLab IMS - Intern Management System

- Designed and deployed a full-stack Intern Management System in production, with a FastAPI backend on AWS EC2 and a React frontend on AWS Amplify, serving live users at careers.looplab.lk
- Built async backend architecture using SQLAlchemy (async) and asyncpg with PostgreSQL on AWS RDS, including database schema design and Alembic migrations
- Implemented RESTful APIs with structured authentication flows and file storage integration via AWS S3
- Managed end-to-end infrastructure including systemd service deployment, Nginx reverse proxy configuration, and SSL via Certbot
- Tools Used: FastAPI, React, PostgreSQL, AWS EC2, AWS RDS, AWS S3, AWS Amplify, SQLAlchemy, Alembic, Nginx

AI-Powered Hospital Management System

- Developed an AI-powered Hospital Management System integrating predictive analytics for inventory, equipment maintenance, and patient load forecasting
- Built four machine learning engines for drug expiry prediction, inventory consumption forecasting, equipment maintenance analysis, and patient traffic forecasting
- Tools Used: React, FastAPI, PostgreSQL, CatBoost, Scikit-Learn, AWS EC2, Tailwind CSS

Cancer Detective PRO

- Built a Django and React web app for early detection of skin, lung, and leukemia cancers using 30K+ medical images, achieving up to 98% detection accuracy
- Containerized FastAPI-based ML inference APIs with Docker and deployed on AWS Lambda for scalable real-time predictions

EDUCATION

Horizon Campus — Malabe, Sri Lanka
BSc (Hons) in Information Technology — May 2021 - Oct 2025
CGPA: 3.30/4.00 | Second Class Upper Division Honours
Relevant Coursework: Software Engineering, Operating Systems, Cloud Computing, Artificial Intelligence, Data Science

REFEREES

Mrs. Thameera Edirisinghe
Manager - Software Development, Digital Platforms
Sri Lanka Telecom PLC
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